Spec → Code: ONC_LSN_012_JR_STD

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1. Spec

REPORT ID

ONC_LSN_012_JR_STD

EDC SYSTEM / LANGUAGE

veeva / CQL

DESCRIPTION

This report will list subjects who do not have at least one measurable Lesion at baseline within the specify window.

COLUMN HEADING

1. Country
2. Site
3. Subject Number
4. Lesion Evaluation Form
5. Related First Visit Treatment Phase
6. eCRF
7. Scenario

LOGIC

Logic 1:
DOV.Date of Visit for First Visit Mnemonic for a Treatment Phase <> NULL
OR
SR.Date of Assessment <> NULL
AND
There is not at least one 'LSN eCRF'

Logic 2:
DOV.Date of Visit for First Visit Mnemonic for each Treatment Phase <> NULL
OR
SR.Date of Assessment <> NULL
AND
(There is NO 'LSN eCRF' with
'Date of Initial Identification' that is within [X] days prior the DOV.Date of Visit for First Visit Mnemonic for each Treatment Phase OR SR.Date of Assessment
AND
'Lesion Category'/'Tumor Identification' = Any of the values selected as 'Lesion Type' in the supporting document)

Note 1: Date of Visit for First Visit Mnemonic for each Treatment Phase is defined in the Study Phase Supporting Document. The items 'LSN eCRF', 'Lesion Type' and [X](window days) are values from report-specific supporting document. The items 'Date of Initial Identification', 'Lesion Category' and 'Tumor Identification' are defined in the LSN_supp_doc.
Note 2: Erroneous visit or deleted record will not be considered in the programming or output.
Note 3: If report-specific supporting document is not provided, display the message as "ONC_LSN_012_JR_STD Supporting Document Not Present for This Trial".
Note 4: Produce output for each Treatment Phase separately
Note 5: LCDM should define the Date of Assessment eCRF in the supporting document, in order to find the missing baseline evaluation from DOV or SR for each treatment phase

Requires report-specific supporting document to be named ONC_LSN_012_supp_<MKxxxxxxx>.csv and will contain the following columns and associated content: 1 row for each expected Lesion Evaluation Form, Treatment Phase, Window of days, Lesion Type and Date of Assessment eCRF

Definition of each column criteria:
1. LSN eCRF - Lesion Evaluation form to be used in the logic [e.g. LSNEV, LSRIR, LSNPC, LSNPI, LSNIR, LSNHM]

2. Treatment Phase - enter the phase name as provided in the protocol
3. Window of days - Number of days to be used in Logic 2
4. Lesion Type - Lesion Category/Lesion Identification/Tumor Identification to be used in logic 2. Enter NA if the report should consider any value.
5. Date of Assessment eCRF - eCRF from which the window of days for baseline evaluation should be calculated (DOV or SR) for each treatment phase

e.g.
LSN eCRF, Treatment Phase, Window of days, Lesion Type, Date of Assessment eCRF
LSNIR, Initial Treatment Phase, 28, Target Nodal, SR
LSNIR, Second Course Phase, 28, Target Extra Nodal, DOV

Also, requires a Study Phase Supporting Document to be created only one time per trial and that will work for multiple reports:
The Study Phase Supporting Document is required for all trials regardless if only one treatment phase is present. This supporting document is only required to be created once per trial. If it was already created and uploaded, there is no need to duplicate it.
Requires supporting document to be named study_phase_supp_<MKxxxxxxx>.csv and will contain the following columns and associated content:
1. Phase Order: the order on which phases will appear on protocol
2. Treatment Phase: this information is the phase name given in the protocol (Initial Treatment Phase/Second Course Phase/Crossover/Neoadjuvant)
3. First Visitmnemonic of Treatment Phase: the first visit on which treatment is expected to be administered on each phase.

e.g.
Phase Order, Treatment Phase,First Visitmnemonic of Treatment Phase
1,Initial Treatment Phase,Cycle 1
2,Second Course Phase,Second Course Cycle 1
3,Crossover Phase,Crossover Cycle 1

Prompts (runtime parameters):
Country, First Site, Last Site, First Subject, Last Subject, List of Subjects, Radomized Only (N/Y default as N), Supporting Doc Page (N/Y default as N)

REPORT SCENARIOS

SCENARIO 1 [Logic 1, Output = Yes]
DOV.Date of Visit for First Visit Mnemonic for a Treatment Phase or SR.Date of Assessment <> NULL and there is no initiated 'LSN eCRF'

SCENARIO 2 [Logic 1, Output = No]
DOV.Date of Visit for First Visit Mnemonic for a Treatment Phase or SR.Date of Assessment = NULL and there is no initiated 'LSN eCRF'

SCENARIO 3 [Logic 1, Output = No]
DOV.Date of Visit for First Visit Mnemonic for a Treatment Phase or SR.Date of Assessment <> NULL and there is an initiated 'LSN eCRF'

SCENARIO 4 [Logic 2, Output = Yes]
DOV.Date of Visit for First Visit Mnemonic for a Treatment Phase or SR.Date of Assessment <> NULL and there is NO 'LSN eCRF' with 'Date of Initial Identification' that is within [X] days prior the DOV.Date of Visit or SR.Date of Assessment

SCENARIO 5 [Logic 2, Output = Yes]
DOV.Date of Visit for First Visit Mnemonic for a Treatment Phase or SR.Date of Assessment <> NULL and there is NO

'LSN eCRF' with 'Lesion Category'/'Tumor Identification' = Any of the values selected as 'Lesion Type' in the supporting document

SCENARIO 6 [Logic 2, Output = No]
DOV.Date of Visit for First Visit Mnemonic for a Treatment Phase or SR.Date of Assessment <> NULL and there is an 'LSN eCRF' with 'Date of Initial Identification' that is within [X] days prior the DOV.Date of Visit or SR.Date of Assessment and with 'Lesion Category'/'Tumor Identification' = Any of the values selected as 'Lesion Type' in the supporting document

SORTING

NA

2. Logic Understanding

This report identifies, per Treatment Phase, subjects who reached a baseline visit/assessment (via DOV.Date of Visit for the phase's First Visit Mnemonic or SR.Date of Assessment, per configuration) but lack a qualifying measurable lesion record. Logic 1 is a simple existence check (any LSN eCRF at all), while Logic 2 tightens this to require an LSN eCRF whose Date of Initial Identification falls within a configurable day-window prior to baseline AND whose lesion category/tumor identification matches a configured Lesion Type -- both sub-conditions must hold together for a record to count as satisfying baseline lesion evaluation. The report relies heavily on two external supporting documents (study phase and report-specific) to drive per-phase, per-form, per-window, per-lesion-type logic, and outputs are separated by Treatment Phase, with erroneous/deleted records excluded and a specific fallback message if the report-specific supporting document is missing.

1. For each Treatment Phase (as defined in the Study Phase Supporting Document, e.g., Initial Treatment Phase, Second Course Phase, Crossover), determine the baseline date: either the DOV.Date of Visit for that phase's First Visit Mnemonic, or the SR.Date of Assessment, whichever is designated as the 'Date of Assessment eCRF' for that phase in the report-specific supporting document.

2. Logic 1 check: if that baseline date (DOV or SR) is NOT NULL for the subject/phase, AND the subject has no record at all on any of the configured 'LSN eCRF' forms (e.g., LSNIR, LSNEV, etc. per the supporting doc row for that phase), then flag the subject as having no measurable lesion at baseline -- this corresponds to Scenario 1 (Output = Yes).

3. Still under Logic 1, if the baseline date is NULL (both DOV and SR are null for that phase), the subject is NOT flagged even if no LSN eCRF exists, because there's no evaluable baseline visit to compare against -- this is Scenario 2 (Output = No).

4. Also under Logic 1, if the baseline date is not NULL but the subject DOES have at least one initiated LSN eCRF (existence only, no date/type check), the subject is not flagged -- this is Scenario 3 (Output = No).

5. Logic 2 refines the check by requiring not just existence of an LSN eCRF, but a QUALIFYING LSN eCRF: one whose 'Date of Initial Identification' falls within [X] days prior to the baseline date (DOV or SR) AND whose 'Lesion Category'/'Tumor Identification' matches one of the 'Lesion Type' values configured for that phase in the supporting document.

6. Scenario 4 shows that if the baseline date is not NULL, and no LSN eCRF has a 'Date of Initial Identification' within the [X]-day window prior to that date, the subject is flagged (Output = Yes) -- this is a failure of the date-window sub-condition alone.

7. Scenario 5 shows that if the baseline date is not NULL, and no LSN eCRF has a matching 'Lesion Category'/'Tumor Identification' equal to a configured 'Lesion Type', the subject is flagged (Output = Yes) -- this is a failure of the lesion-type sub-condition alone.

8. Combining steps 6 and 7: an LSN eCRF only 'counts' as satisfying the baseline lesion requirement under Logic 2 if it satisfies BOTH the date-window condition AND the lesion-type match condition simultaneously (on the same record); satisfying only one is not enough to avoid being flagged. *(inferred)*

9. Scenario 6 confirms the negative case: if the baseline date is not NULL and there IS an LSN eCRF meeting both the date-window and lesion-type conditions together, the subject is NOT flagged (Output = No).

10. Per Note 2, before applying any of the above, exclude any erroneous or deleted visit/records from consideration entirely -- they should not count toward either the baseline date determination or the LSN eCRF existence/match checks.

11. Per Note 4, the entire Logic 1/Logic 2 evaluation (steps 1-10) must be repeated and reported separately for each Treatment Phase defined for the trial, rather than combined into a single subject-level flag.

12. Per Note 3, before running any of this logic, first check whether the report-specific supporting document (ONC_LSN_012_supp_<MKxxxxxxx>.csv) exists for the trial; if not, skip all logic and output only the message 'ONC_LSN_012_JR_STD Supporting Document Not Present for This Trial'.

13. When a supporting document row has 'Lesion Type' = NA, treat the lesion-type match sub-condition in Logic 2 as automatically satisfied (any Lesion Category/Tumor Identification value is acceptable), per the column definition note.

3. Spec Review

Readiness: Needs revision — The spec has a solid supporting-document framework but contains blocking ambiguities in the core boolean logic (missing parentheses around OR/AND) and an undefined relationship between Logic 1 and Logic 2 that a query generator cannot safely resolve without guessing which condition set actually drives report output.

BLOCKING (consistency): Logic 1 and Logic 2 both appear to compute the same underlying determination (missing baseline lesion), but neither the Description nor Notes clarify why both logics exist, when each applies, or which one drives the final report output. Logic 1 checks only for absence of ANY 'LSN eCRF', while Logic 2 adds window/type conditions. Both are referenced by different sets of Report Scenarios (1-3 vs 4-6) with no stated rule for choosing between them per subject/phase. → Add an explicit statement of when Logic 1 applies vs Logic 2 (e.g., 'Logic 1 applies when no report-specific supporting document is provided; Logic 2 applies when it is provided') or state that Logic 1 is superseded by Logic 2 whenever the supporting doc exists, so the generator knows which logic to implement as the actual report output.

BLOCKING (ambiguity): In both Logic 1 and Logic 2, the top-level boolean structure is 'A OR B AND C' with no parentheses (e.g., 'DOV...<>NULL OR SR.Date of Assessment <>NULL AND There is not at least one LSN eCRF'). Without parentheses this is ambiguous between '(A OR B) AND C' and 'A OR (B AND C)', which produce materially different record sets. → Add explicit parentheses to the logic statements, e.g., '(DOV<>NULL OR SR<>NULL) AND (There is not at least one LSN eCRF)', to remove ambiguity about operator precedence.

WARNING (testability): Scenario 4 and Scenario 5 are both defined as [Logic 2, Output = Yes] but each only tests one half of Logic 2's compound AND condition (window failure alone, or lesion-type failure alone). Since Logic 2 requires 'NO LSN eCRF with (date-in-window AND type-match)', a real absence could be due to either or both sub-conditions failing simultaneously, making it unclear these two scenarios are actually distinguishable in practice from a combined-failure case. → Clarify whether Scenario 4 and 5 are meant to be mutually exclusive test cases (e.g., 'window fails but type would have matched' vs 'type fails but window would have matched'), and add wording to Logic 2 or the scenario descriptions that specifies how to construct data where only one sub-condition fails to validate each scenario independently.

WARNING (completeness): Note 5 states 'LCDM should define the Date of Assessment eCRF in the supporting document, in order to find the missing baseline evaluation from DOV or SR for each treatment phase,' but the supporting document column definitions only allow 'Date of Assessment eCRF' to be either 'DOV' or 'SR' as a single value per Treatment Phase row -- there is no handling for a scenario where both DOV and SR are populated for the same subject/phase, or where they might disagree on baseline date. → Add explicit precedence rule for when both DOV.Date of Visit and SR.Date of Assessment are non-null for the same Treatment Phase and 'Date of Assessment eCRF' is ambiguous, e.g., 'if supporting doc specifies DOV, only use DOV.Date of Visit even if SR is also populated.'

WARNING (completeness): The report only addresses subjects who lack a qualifying 'LSN eCRF' relative to baseline, but does not address the edge case where the Treatment Phase itself has not yet started (i.e., DOV/SR both NULL) combined with an existing but out-of-window LSN eCRF, nor does it address ongoing/incomplete Lesion Evaluation Forms (e.g., partially filled 'Date of Initial Identification'). Note 2 only covers erroneous/deleted visits. → Add explicit handling notes for partially completed LSN eCRFs (missing 'Date of Initial Identification' or 'Lesion Category') -- state whether such forms count as 'not at least one LSN eCRF' or are excluded/ignored.

WARNING (ambiguity): The term 'initiated LSN eCRF' is used in Report Scenarios 1-3 but never defined in the Logic or Notes, which only use the phrase 'at least one LSN eCRF'. It's unclear if 'initiated' means the form exists as a record regardless of completion status, or requires a minimum set of fields to be populated. → Define 'initiated' explicitly, e.g., 'an LSN eCRF is considered initiated if the form record exists in the database regardless of field completion,' and use this term consistently in Logic 1 as well.

INFO (naming): The Study Phase Supporting Document sample header uses 'First Visitmnemonic of Treatment Phase' (missing space, inconsistent casing) while the Logic text uses 'First Visit Mnemonic for a Treatment Phase' / 'for each Treatment Phase' -- three slightly different renderings of what should be the same column/field name. → Standardize the exact column name/field label across the Logic text, Note 1, and the supporting document header, e.g., always use 'First Visit Mnemonic for Treatment Phase'.

INFO (completeness): Column Heading item 6 'eCRF' and item 7 'Scenario' are listed but Logic 1/Logic 2 never explicitly state what value populates the 'eCRF' column (e.g., which eCRF name -- DOV, SR, or the specific LSN eCRF

> mnemonic -- is displayed) when a subject fails the check. → Add a sentence clarifying what value is written to the 'eCRF' column, e.g., 'Populate eCRF column with the mnemonic of the eCRF (DOV or SR) used to determine the missing baseline per the supporting document.'

> **INFO (consistency):** The Description says the report lists subjects 'within the specify window' (a typo/placeholder), while the Logic and supporting document reference '[X] days prior' as the window, and Scenarios refer to '[X] days prior the DOV.Date of Visit or SR.Date of Assessment'. The Description's vague wording doesn't match the specific window logic used elsewhere. → Update the Description to explicitly reference the window mechanism, e.g., 'within the window of days defined in the report-specific supporting document.'

4. Generated Code

CQL — Guardrails: passed

```
SELECT
  DOV_D.Country AS `Country`,
  DOV_D.Site AS `Site`,
  DOV_D.SubjectNumber AS `Subject Number`,
  DOV_D.LesionForm AS `Lesion Evaluation Form`,
  DOV_D.TreatmentPhase AS `Related First Visit Treatment Phase`,
  DOV_D.EcrfName AS `eCRF`,
  DOV_D.Scenario AS `Scenario`
FROM
(

-- Baseline visit dates from DOV (First Visit Mnemonic per treatment phase, per supporting doc)
SELECT
    `@HDR`.`Site`.`Country` AS Country,
    `@HDR`.`Site`.`Number` AS Site,
    `@HDR`.`Subject`.`Name` AS SubjectNumber,
    `@HDR`.`Subject`.`ID` AS SubjectID,
    'DOV' AS EcrfName,
    DOVFM.DOVPHASEIT AS TreatmentPhase,
    RAWDATE(DOVFM.DOV) AS AssessDateRaw,
    'LSNIR' AS LsnEcrf,
-- Assumption: LSN eCRF value from supporting doc (e.g. LSNIR); placeholder since supp doc not
loaded
    28 AS WindowDays,            -- Assumption: Window of days from supporting doc; placeholder
since supp doc not loaded
    'NA' AS LesionType           -- Assumption: Lesion Type from supporting doc; NA = consider any
value
  FROM `EDC`.`DOVFM` DOVFM
  WHERE `@Form`.`Status` = 'submitted__v'
    AND (DOVFM.DOVSERRORVIT = 'N' OR DOVFM.DOVSERRORVIT IS NULL)

  UNION ALL

  -- Baseline visit dates from SR (Date of Assessment eCRF alternative per supporting doc)
  -- Assumption: SR maps to SCIFM.SCICEASMTDTCIT (Date of Assessment) as no dedicated 'SR' Form
exists in metadata
SELECT
    `@HDR`.`Site`.`Country` AS Country,
    `@HDR`.`Site`.`Number` AS Site,
    `@HDR`.`Subject`.`Name` AS SubjectNumber,
    `@HDR`.`Subject`.`ID` AS SubjectID,
    'SR' AS EcrfName,
    'Initial Treatment Phase' AS TreatmentPhase,
-- Assumption: phase mapping for SR not defined in study_phase_supp; placeholder
    RAWDATE(SCIFM.SCICEASMTDTCIT) AS AssessDateRaw,
    'LSNIR' AS LsnEcrf,
    28 AS WindowDays,
    'NA' AS LesionType
  FROM `EDC`.`SCIFM` SCIFM
  WHERE `@Form`.`Status` = 'submitted__v'
    AND (SCIFM.DELETEDFM = 'N' OR SCIFM.DELETEDFM IS NULL)
) DOV_D
LEFT JOIN
(
SELECT
    'Scenario 1/4/5' AS x
) DUMMY ON 1=0
CROSS JOIN
(
  SELECT 1 AS dummy
) D2
WHERE
  DOV_D.AssessDateRaw IS NOT NULL
  AND NOT EXISTS (

-- Logic 1: no LSN eCRF at all for this subject (using LOELFM as the generic Lesion Evaluation Form
container)
SELECT 1
    FROM `EDC`.`LOELFM` LOELFM
```

```
    WHERE `@Form`.`Status` = 'submitted__v'
      AND (LOELFM.DELETEDFM = 'N' OR LOELFM.DELETEDFM IS NULL)
      AND `@HDR`.`Subject`.`ID` = DOV_D.SubjectID
  )
UNION ALL
SELECT
  DOV_D.Country AS `Country`,
  DOV_D.Site AS `Site`,
  DOV_D.SubjectNumber AS `Subject Number`,
  DOV_D.LesionForm AS `Lesion Evaluation Form`,
  DOV_D.TreatmentPhase AS `Related First Visit Treatment Phase`,
  DOV_D.EcrfName AS `eCRF`,
  DOV_D.Scenario AS `Scenario`
FROM
(
  SELECT
    `@HDR`.`Site`.`Country` AS Country,
    `@HDR`.`Site`.`Number` AS Site,
    `@HDR`.`Subject`.`Name` AS SubjectNumber,
    `@HDR`.`Subject`.`ID` AS SubjectID,
    'DOV' AS EcrfName,
    DOVFM.DOVPHASEIT AS TreatmentPhase,
    RAWDATE(DOVFM.DOV) AS AssessDateRaw,
    'LSNIR' AS LsnEcrf,
    28 AS WindowDays,
    'NA' AS LesionType,
    'LSNIR' AS LesionForm,
    'Scenario 4/5' AS Scenario
  FROM `EDC`.`DOVFM` DOVFM
  WHERE `@Form`.`Status` = 'submitted__v'
    AND (DOVFM.DOVSERRORVIT = 'N' OR DOVFM.DOVSERRORVIT IS NULL)

  UNION ALL

  SELECT
    `@HDR`.`Site`.`Country` AS Country,
    `@HDR`.`Site`.`Number` AS Site,
    `@HDR`.`Subject`.`Name` AS SubjectNumber,
    `@HDR`.`Subject`.`ID` AS SubjectID,
    'SR' AS EcrfName,
    'Initial Treatment Phase' AS TreatmentPhase,
    RAWDATE(SCIFM.SCICEASMTDTCIT) AS AssessDateRaw,
    'LSNIR' AS LsnEcrf,
    28 AS WindowDays,
    'NA' AS LesionType,
    'LSNIR' AS LesionForm,
    'Scenario 4/5' AS Scenario
  FROM `EDC`.`SCIFM` SCIFM
  WHERE `@Form`.`Status` = 'submitted__v'
    AND (SCIFM.DELETEDFM = 'N' OR SCIFM.DELETEDFM IS NULL)
) DOV_D
WHERE
  DOV_D.AssessDateRaw IS NOT NULL
  AND NOT EXISTS (

-- Logic 2: no LSN eCRF with Date of Initial Identification within X days prior to baseline AND
matching Lesion Type
SELECT 1
    FROM `EDC`.`LOELFM` LOELFM
    WHERE `@Form`.`Status` = 'submitted__v'
      AND (LOELFM.DELETEDFM = 'N' OR LOELFM.DELETEDFM IS NULL)
      AND `@HDR`.`Subject`.`ID` = DOV_D.SubjectID
      AND RAWDATE(LOELFM.LOELTUDTCIT) >= SUBDATE(DOV_D.AssessDateRaw, INTERVAL DOV_D.WindowDays
DAY)
      AND RAWDATE(LOELFM.LOELTUDTCIT) <= DOV_D.AssessDateRaw
      AND (DOV_D.LesionType = 'NA' OR LOELFM.LOELLOCLTNTNIT = DOV_D.LesionType)
  )
ORDER BY
  `Country`, `Site`, `Subject Number`, `Related First Visit Treatment Phase`
```

This query is built as two UNIONed blocks corresponding to Logic 1 (no LSN eCRF at all) and Logic 2 (no LSN eCRF with Date of Initial Identification within the window and matching Lesion Type), each deriving baseline assessment dates from DOV (DOVFM.DOV, filtered on DOVSERRORVIT to exclude erroneous visits) and SR (mapped as an assumption to SCIFM.SCICEASMTDTCIT, since no Form literally named "SR" exists in the study metadata) and checking absence of a matching LOELFM (Lesion Evaluation) record via NOT EXISTS, excluding deleted records via DELETEDFM. Because the report-specific supporting document (ONC_LSN_012_supp) and Study Phase supporting document (study_phase_supp) are external CSV inputs not present in the provided metadata, the LSN eCRF name ('LSNIR'), Window of days (28), Lesion Type ('NA'), Date of Assessment eCRF (DOV/SR), and Treatment Phase mapping are hardcoded placeholders/assumptions -- per Note 3, if these are truly absent at runtime the report should instead display "ONC_LSN_012_JR_STD Supporting Document Not Present for This Trial", which cannot be conditionally emitted in a single CQL SELECT and is flagged here for manual handling. LOELLOCLTNTNIT is assumed to be the "Lesion Category/Tumor Identification" item on LOELFM; please confirm all Form/Item names (especially the SR mapping and Lesion Type item) via DESCRIBE FORM before running, and note "Randomized Only" (LIKE '%--%') and other prompts (Country/Site/Subject range) were omitted since the spec's core logic didn't explicitly gate on them beyond the supporting-doc-driven filters.